

chap_simples_SY

2025-07-09

PLAN SY

```
a <- 13/3
(u <- runif(1,min=0,max=a))

## [1] 0.7801507
# [1] 3.337597
u+0:2*13/3

## [1] 0.7801507 5.1134840 9.4468173
# [1] 3.337597 7.670930 12.004263
ceiling(u+0:2*a)

## [1] 1 6 10
# [1] 4 8 13

library(sampling)
pik <- rep(3/20, 20)
UPsystematic(pik)

## [1] 0 0 0 0 0 1 0 0 0 0 0 0 1 0 0 0 0 0 0 1
# [1] 0 1 0 0 0 0 0 1 0 0 0 0 0 0 1 0 0 0 0 0

library(sampling)
data(rec99)
N <- nrow(rec99)
n <- 70
pik <- rep(n/N,N)
set.seed(123456)
sy.rec <- UPsystematic(pik)

library(survey)

## Loading required package: grid
## Loading required package: Matrix
## Loading required package: survival
##
## Attaching package: 'survival'
## The following objects are masked from 'package:sampling':
##
##   cluster, strata
##
```

```
## Attaching package: 'survey'

## The following object is masked from 'package:graphics':
##
##      dotchart

ech.sy <- svydesign(id=~CODE_N, weights=rep(N/n,n),
fpc=rep(n/N,n), data=rec99[sy.rec==1,])
svytotal(~LOGVAC, ech.sy)

##          total    SE
## LOGVAC 14998 2668

# total
# SE
# LOGVAC 7795.6 980.08
```